Biochemistry Cumulative Exam

Cytochrome oxidase is an integral membrane protein in the mitochondrial respiratory chain which transfers four electrons from four molecule of reduced cytochrome c to one molecule of O_2, and pumps 4 protons from the inside to the outside of the mitochondrial membrane, and uses four additional protons from inside the membrane to form two water molecules.

$$4 \text{ cyt c (reduced)} + O_2 + 8\ H^+_{in} \rightarrow 4 \text{ cyt c (oxidized)} + 2\ H_2O + 4\ H^+_{out}$$

Cytochrome c has a redox potential of $E_o' = +0.26$ V and the $O_2/2H_2O$ reaction has a redox potential of +0.82. Under active electron transfer conditions, the electrical potential across the mitochondrial membrane was measured to be $V_{out} - V_{in} = +0.25$ V, and the pH inside and outside the mitochondrial membrane was $pH_{in} = 8.5$, and $pH_{out} = 6.0$.

1. (30) Draw a diagram of cytochrome oxidase showing the pathway of electron transfer from cytochrome c through all the redox centers of cytochrome oxidase to oxygen. Give a mechanism for the reduction of oxygen to water catalyzed by cytochrome oxidase, giving the key intermediates. Give a diagram showing how cytochrome oxidase couples the energy of each step in the above mechanism to pumping protons across the mitochondrial membrane, and discuss the mechanism of proton pumping.

2. (30) Calculate the free energy change ΔG^o for electron transfer from 4 moles of reduced cytochrome c to one mole of O_2 . (not including the energy of proton pumping).

3. (30) Calculate ΔG for the proton pumping component of the cytochrome oxidase reaction under the above conditions.

4. (10) Calculate how efficiently cytochrome oxidase couples the exergonic free energy of electron transfer to the endergonic free energy of proton pumping under the above conditions.

Possibly useful formulas:

For the reaction: $aA + bB \rightleftarrows cC + dD$; $\Delta G = \Delta G^{0'} + RT\ln\{[C]^c[D]^d/[A]^a[B]^b\}$

$R = 0.00831\ kJmol^{-1}K^{-1}$ $T = 298\ ^oK$ at $25\ ^oC$ $\log_{10}C = 2.303 \ln C$

$\Delta G = \Delta G^{o'} + 5.70 \log_{10}\{[C]^c[D]^d/[A]^a[B]^b\}$kJ/mol.

At equilibrium: $\Delta G = 0$, $\Delta G^{0'} = -RT\ln\{K_{eq}\}$ J/mol $= -5.70 \log_{10}\{K_{eq}\}$ kJ/mol;

$K_{eq} = \{[C]^c[D]^d/[A]^a[B]^b\}$ at equilibrium. Use molar units.

For redox reactions: $\Delta G^{o'} = -nF\Delta E^{o'}$ n = # electrons, Faraday's constant $F = 96.5$ kJ/volt.mol

For the transport of 1 mole of H^+ from inside to outside a membrane:

$\Delta G(H^+_{in} \rightarrow H^+_{out}) = 5.7\log([H^+_{out}]/[H^+_{in}]) + F\ (V_{out} - V_{in}) = 5.7(pH_{in} - pH_{out}) + F\ (V_{out} - V_{in})$